Patient Churn Analysis — Tableau Dashboarding (Descriptive Phase)

Dataset Name: Medical Appointment No Shows Dataset
Dataset Used: patient_churn.csv
Date: 29-04-2026

Introduction:

This report presents the development of a descriptive Tableau dashboard for analyzing patient churn in a healthcare setting. The dataset includes patient demographics, appointment details, and service-related attributes. The primary objective of this phase is to visualize current trends and patterns related to patient no-shows and engagement behavior using an interactive dashboard.

Dataset Justification:

The Medical Appointment dataset was selected for this task due to the following reasons:

- It represents real-world healthcare data, making the analysis practical and relevant.
- It contains both categorical and numerical features, enabling meaningful visualizations.
- The dataset includes a clear target variable (No-show), which represents patient churn.
- It supports analysis of patient behavior through attributes such as age, gender, and appointment details.

Task Objective:

- Connect Tableau to the cleaned dataset
- Build a “Current State” executive dashboard
- Analyze overall patient churn rate
- Identify revenue at risk due to missed appointments
- Explore demographic and behavioral patterns
- Implement filters and tooltips for interactivity

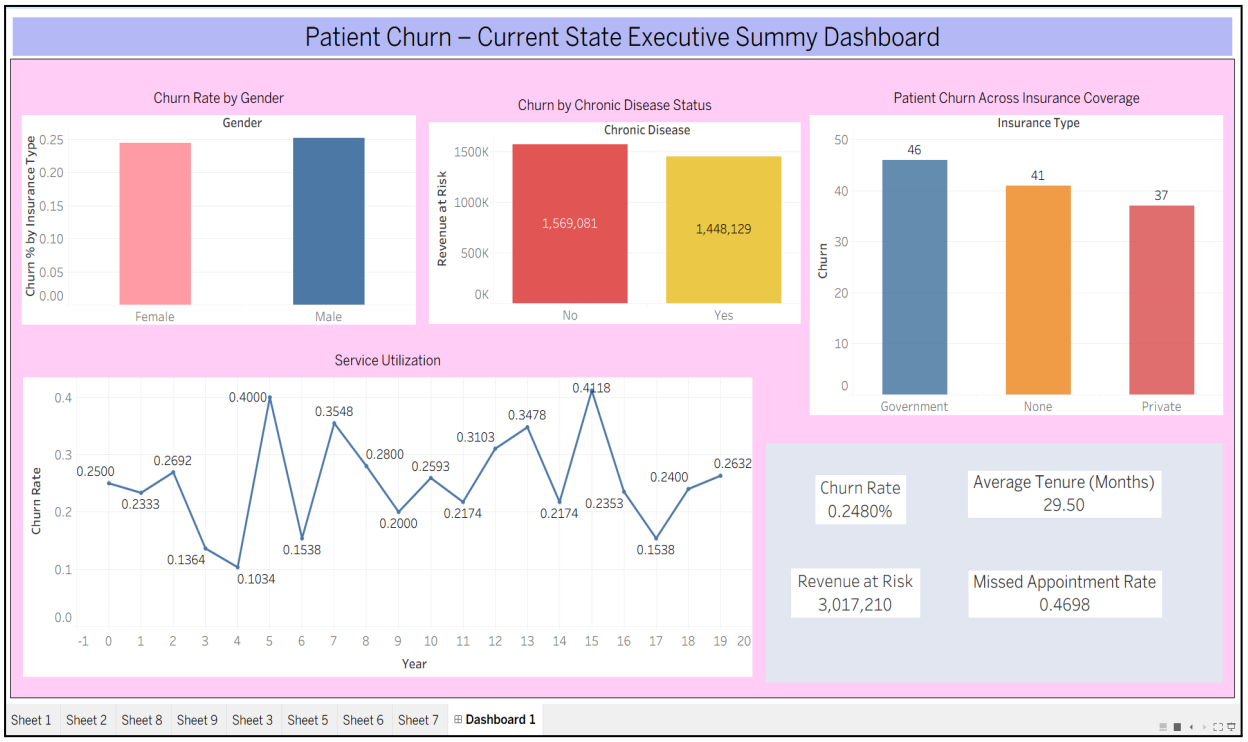
Tools Used:

- Tableau

Task Performed:

- Imported the cleaned healthcare dataset into Tableau and verified data types.
- Created calculated fields such as Patient Churn Flag, Churn Rate, and Revenue at Risk using conditional logic.
- Designed KPI cards to display key metrics including churn rate, average tenure, revenue at risk, and missed appointment rate.
- Developed multiple visualizations including:
    - Churn rate by gender
    - Revenue at risk by chronic disease status
    - Patient churn across insurance coverage
    - Service utilization trends over time
- Applied color coding to differentiate churned and retained patients for better visual clarity.
- Added dynamic filters for key variables such as gender, insurance type, and appointment characteristics.
- Customized tooltips to provide detailed insights on hover, including patient count and churn percentages.
- Structured the dashboard layout to ensure readability and effective presentation of insights.

Dashboard:



Dashboard Description:

The Patient Churn – Current State Executive Summary Dashboard presents a clear overview of patient engagement within a healthcare system, with a focus on identifying patterns related to missed appointments. It brings together key indicators such as overall churn rate, revenue at risk, average tenure, and missed appointment rate to provide a quick understanding of performance.

The dashboard explores how churn varies across different segments, including gender, insurance coverage, and chronic disease status. It also includes a service utilization trend to observe changes in patient behavior over time. These visual elements help highlight areas where patient drop-off is more frequent.

Interactive components such as filters and tooltips allow users to refine the analysis and view detailed information instantly. The layout is designed to be simple and structured, making it easy to interpret insights and support informed decision-making in a healthcare environment.

Conclusion:

The Tableau dashboard successfully provides a descriptive overview of patient churn in the healthcare domain. It highlights key metrics and uncovers patterns related to patient behavior, demographics, and service utilization. The interactive design allows users to explore data dynamically, making it easier to identify high-risk segments and potential areas for improvement. This dashboard serves as a strong foundation for further predictive analysis and decision-making in healthcare management.